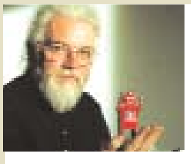
ing an 'energy-sucking fang' into a specific spot on them to transfer battery power to themselves. This may sound a little grey, but it's as simple as that. At Rotherham, predator and prey robots struggled for survival, feeding and hunting, with their creators hoping they would succeed in building a civilisation. How would they evolve? Quite simply, by the Darwinian principle: each bot would upload its genes, when they changed, to a central server. And in the next generation, only those bots that lived long enough would have their genes added to the gene pool.

But there's more. In June 2002, one of the bots went missing. It had escaped. It had left the arena for a small repair to its body work after a day of particularly hard battling. It forced its way through a small gap in the barriers, went down the access slope to the workshop and out into the car park. It somehow made its way to the entrance gates, only to be found by an unassuming visitor who nearly ran over it in his car.

Noel Sharkey, the brain behind the Magna experiment, said he was thrilled that the robots have become so intelligent. He said it was fantastic and, "there's no need to worry, as although they can escape, they are perfectly harmless and won't be taking over just yet!"

However, in a cyberchat with the BBC, when asked whether there would be a time when we wouldn't be able to distinguish a human from a bot, Professor Sharkey said "It's unlikely that anything we think of as a robot now will be indistinguishable from a human. I believe that metal will never be mental."



Noel Sharkey
Interdisciplinary researcher
University of Sheffield, England

"The name of the game is to develop extremely simple mechanisms that, when placed in the world, exhibit complex behaviour."

Self-modifying software and hardware

News about robots that can play with your dog, or set up a LAN, is boring. We've come to expect robots to be able to do stuff like this. And after experiments like the one at Magna, we expect them to be even smarter. In the fundamental shift that will take place, robots will be able to know about themselves, and make changes to their software and hardware.

Jürgen Schmidhuber, a major contributor to the field of learning robots, talks about Gödel machines—how a machine will modify its software. A robot powered by a Gödel machine will be able to modify its code, as soon as it proves that its performance will improve with the modified code. As an example application of Gödel machines, Schmidhuber mentions a robot that interacts with a partially unknown environment, trying to find hidden gasoline depots to occasionally refuel its tank. Such a robot, if it had a Gödel machine for a controller, would be able to optimise itself so that it lives the longest it can, by refuelling itself optimally.

Self-modifying software is one thing, and self-modifying hardware is quite another. This is generally called 'evolvable hardware', and there are several research groups working towards the fabled robot that opens up a body part and repairs it when something goes wrong.

Artificial evolution was previously relegated to software simulations, carried out on neural network configurations, and that was considered enough. But as Adrian Thompson of the University of Sussex—a pioneer in the field of Evolutionary Robotics (ER)—points out, "There are three reasons why we would want hardware to modify itself. First, hardware evolution can deal with real-world physics, which is difficult to analyse or model in simulations. Second, when it comes to interacting entities, it is difficult to precisely model size, location, shape and so on, in simulations. And third, the characteristics and interactions of the evolving entities do not remain the same with time—physical evolution will be able to get round this problem."

FPGAs

FPGAs, or Field Programmable Gate Arrays, are programmable digital logic chips. No soldering, or any such manual work is required to change their function. All that is needed is for the FPGA's program, or desired behaviour, to be downloaded onto the chip.
The program for the FPGA, or the function that it will perform, needs to be first designed on a computer, and then translated into a form understandable by the FPGA—that is, it needs to be compiled. The compiled program is downloaded onto the FPGA, and it then behaves according to the design. For example, you can program an FPGA to perform a basic logic operation, such as 'And'. Of course, they can be programmed to perform much more complex operations.

Bots changing themselves

So what exactly does evolving hardware look like? Not like a humanoid wielding a screwdriver—at least not yet. The Khepera robot (***http://www.k-team.com/robots/khepera/***) has been modified to have two layers of FPGAs *(See box 'FPGAs' above)* on top, which allow evolution of the robot's control system. FPGAs are hardware that actually changes as the robot evolves.

Evolution can be used to find novel circuit designs by searching a larger 'design space' than is accessible to humans. The evolution of circuits has led researchers to conclude that they could learn from it—they could discover design principles from evolved hardware.

An adaptive hardware system that learned the behaviour of an expert robot controller, using a genetic algorithm, has been developed. This controller could be used if the primary controller failed. Think of this as a robot developing a spare brain for itself after inspecting its original brain. Parts of robot controllers have also been evolved, which interact with the environment dynamically, in a manner that would not have emerged if traditional design had been used.

Robots have evolved to visually discriminate between objects of different sizes. They have evolved over a period of time to integrate sensory and motor